MILITARY SPECIFICATION

ACOUSTIC LOCATING DEVICE

(For Torpedo Mk 46 Mods)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Acoustic Locating Device, Dwg 2158843 and Dwg 3277485.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for

STANDARDS

FEDERAL

FED-STD-H28	Screw Thread Standards for Federal Service

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (524), Indian Head, MD 20640 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this documents or by letter.

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MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-889	Dissimilar Metals

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND (Code Ident 10001)

Dwg 2158843	Locating Device, Acoustic
Dwg 1579452	Battery, Mercury
Dwg 3277485	Locating Device, Acoustic
Dwg 3277407	Battery Pack, ALD

PUBLICATIONS

NAVAL ORDNANCE SYSTEMS COMMAND

OP 3389	Torpedo Mk 46

(Copies of specifications, standards, drawings, and publications required by manufacturers in connections with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposals shall apply.

PUBLICATIONS

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Rules

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(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4. and 6.).

3.2 Basic requirement.

3.2.1 Materials and treatments of materials and parts. Unless otherwise specified in the contract, all materials and treatments of materials and parts used in the manufacture of the Acoustic Locating Device shall be in accordance with the requirements specified herein and with the applicable documents listed on Drawing 2158843 or 3277485.

3.2.1.1 Selection of materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided herein. AN, JAN, or MS Standard parts shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, and cotter pins may be used, provided they possess suitable properities and are replaceable by the AN, JAN, or MS Standard parts without alteration, and provided the corresponding AN, JAN, or MS Standard part numbers are referenced on the parts list, and if practicable, on the contractor's drawings. In the event there is no suitable corresponding standards part in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements specified herein. All electronic components purchased for use in the acoustic locating device, Drawing 3277485, shall meet the requirements of JANTX Discrete Semiconductors and MIL-STD-883 Level B requirements for integrated circuits.

3.2.1.2 Fungus proof material. Materials that are nutrients for fungi shall not be used unless necessary. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity.

3.2.1.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.2.1.4 Threads. Unless otherwise specified, all threads shall be in accordance with FED-STD-H28.

3.2.2 Interchangeability. All parts or subassemblies having the same manufacturer's part number shall be directly and completely interchangeable with respect to installation and performance.

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3.2.3 Design changes. Unless otherwise specified by the procuring activity, any change in design (which in the opinion of the procuring activity affects performance or reliability of the Acoustic Locating Device) made after completion of first article tests, shall require that a new first article sample (see 4.3.1) manufactured in accordance with the design change by submitted for testing.

3.2.4 Gages. The supplier shall provide whatever gages are necessary and adequate to assure that the material and parts of the Acoustic Locating Device will meet dimensions and requirements shown on the applicable documents. When requested by the supplier, the procuring activity will furnish pertinent drawings of existing Naval final inspection gages for use only in the designing of the supplier's gages. This action does not relieve the supplier of the responsibility for designing and manufacturing the gages needed for the satisfactory fulfillment of the contract requirements.

3.2.5 Inspections and test equipment and facilities. The supplier shall furnish and maintain all necessary inspection and test equipment and facilities and shall provide personnel for performing all acceptance tests specified herein. The test equipment shall be adequate to permit performance of the required acceptance tests for Acoustic Locating Device.

3.3 Performance requirements and physical characteristics.

3.3.1 Activiation of signal. The Acoustic Locating Device shall remain in a static, ready condition (no output) until the unit is electrically activitated by one of two means:

(a) Completion of an external circuit from electrical connector (see FIGURE 1). Electrical completion shall consist of one ohm maximum resistance between pins D and E.

(b) The flooding of salt water (3.1 percent minimum salinity across exposed pins E4 and E5 (see FIGURE 1).

3.3.2 Output signal. When tested in accordance with 4.5.2.6 and 4.7.1.2, the output signal shall be a pulsed output of 30 ±5 msec duration at a repetition rate of 60 ±6 per minute at a frequency of 9.000 ±0.270 kHz.

3.3.3 Output level. When tested in accordance with 4.5.2.6 and mounted in the Mk 85 Mod 1 Exercise Head, the acoustic output pressure level of the Acoustic Locating Device shall be 149 (dB) minimum referred to one micropascal at one meter. This output level shall be achieved over a 40 degree arc in the transverse plane of the exercise head. See FIGURE 2 for the transverse plane location and arc description.

3.3.4 Power supply. The power supply for the Acoustic Locating Device, Drawing 2158843, shall be a self-contained replaceable battery (Drawing 1579452). The power supply for the Acoustic Locating Device, Drawing 3277485, shall be a self-contained rechargeable battery (Drawing 3277407).

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3.3.5 Attitude. The Acoustic Locating Device shall be capable of operating in accordance with all requirements when positioned in any attitude.

3.3.6 Configuration. The envelope dimensions, part identification, mounting, and other external dimensions shall be in accordance with the applicable drawings (2158843 or 3277485).

3.3.7 Weight. The weight of the Acoustic Locating Device shall not exceed 3.5 pounds.

3.4 Environmental requirements. Unless otherwise specified, the Acoustic Locating Device shall be capable of conforming to the performance requirements and physical characteristics of this specification after being exposed to the environments specified in this section.

3.4.1 Vibration (see 6.3.1). The Acoustic Locating Device shall be capable of withstanding the vibration shown in TABLE I.

TABLE I. Vibration schedule.

Condition (see 6.3.4)	Double amplitude (inch) minimum	Acceleration (g's) minimum	Frequency range (cps)
Nonoperating, Ready non-operating, and Ready	0.75	-	5 - 6
	-	1.3	6 - 16
	0.10	-	16 - 24
	-	3.0	24 - 40
	0.036	-	40 - 74
	-	10.0	74 - 500

3.4.2 Shock (see 6.3.3). The Acoustic Locating Device shall be capable of withstanding shock specified in TABLE II.

TABLE II. Shock requirements.

Condition (see 6.3.4)	Peak magnitude (g's)	Minimum Duration (ms)	Acceleration direction
Ready	300	0.5	Aft
	85	0.5	Any
	45	20.0	Forward
Nonoperating	60	8.0	Any

3.4.3 Acceleration (see 6.3.4). The Acoustic Locating Device shall be capable of withstanding, in the ready condition (see 6.3.2.3), accelerations of 35.0 g's in the transverse plane.

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3.4.4 Temperature. The Acoustic Locating Device shall be capable of conforming to the requirements of this specification after being at any temperature between -45 and 140 degrees F in the nonoperating condition (see 6.3.2.1) and while at any temperature between 40 and 140 degrees F in the ready nonoperating (see 6.3.2.2) and ready (see 6.3.2.3) conditions.

3.4.5 Leakage. The fully assembled Acoustic Locating Device shall not leak when submerged in water at any pressure up to 1000 psig.

3.5 Life.

3.5.1 Operational service life. The output signal shall not decay more than six db after five days of continuous, unattended operation when the Acoustic Locating Device is submerged in sea water (3.1 percent minimum salintiy) at a pressure of 1000 psig.

3.6 Identification. The Acoustic Locating Device shall be permanently identified in accordance with MIL-STD-130 and the applicable documents listed on Drawing 2158843 or 3277485.

3.7 Workmanship. The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with the requirements of this specification. All parts used in the assembly shall be thoroughly clean and free of excess materials, chips, and loose or splattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample subject to any qualifications stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection. Unless otherwise specified in the contract. the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the Acoustic Locating Device shall be classified as follows:

(a) First article inspection

(1) First article sample (see 4.3.1)

(2) First article tests (see 4.5)

(b) Periodic production inspection

(1) Periodic production sampling (see 4.3.2)

(2) Periodic production tests (see 4.6)

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(c) Acceptance Inspection

(1) Acceptance sampling (4.3.3)

(2) Acceptance tests and examinations (see 4.7.1)

4.3 Sampling.

4.3.1 First article sample. A first article sample of two Acoustic Locating Devices, shall be subjected to the first article tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the first article sample shall be based on no defects in the sample. Further production of the Acoustic Locating Device by the supplier, prior to approval of the first article sample, shall be at the supplier's risk.

4.3.2 Periodic production sampling. Periodic production tests detailed in 4.6 shall be performed at an activity designated by the procuring activity. Samples shall consist of units that have satisfactorily passed the acceptance tests and examinations detailed in 4.7 and selected as follows:

(a) Step 1. Upon initiation of production, randomly select 1 unit from each group of 10 units. Upon satisfactory compliance of samples selected from three consecutive groups, proceed to step 2.

(b) Step 2. Randomly select 1 unit from each group of 25 units. Upon satisfactory compliance of samples selected from three consecutive groups, proceed to step 3.

(c) Step 3. Randomly select 1 unit from each group of 50 units. Continue with step 3 for the balance of the contract or until a defective unit is found.

4.3.2.1 Defective units. When a defective unit is found in any group the supplier shall:

(a) Determine the cause of the defect.

(b) Determine if the defect is an isolated case.

(c) Take the necessary action to eliminate the defect from units on hand and, if so directed by the procuring activity, from those units previously shipped.

(d) Submit to the procuring activity, for approval, the proposed corrective action to prevent recurrence of the defect.

(e) Return to step 1 (4.3.2.(a)) and proceed as indicated for inspection of units subsequently produced.

4.3.3 Acceptance sampling. The tests and examinations detailed in 4.7.1 shall be conducted on each Acoustic Locating Device to be delivered; therefore, there shall be no sampling plan for acceptance. Defective units shall be rejected.

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4.4 Test conditions. Unless otherwise specified, the Acoustic Locating Device shall be subjected to tests under the following conditions:

(a)	Temperature	Room ambient (60 to 95 degrees F)
(b)	Humidity (Relative)	Room ambeient, up to 95 percent
(c)	Altitude	Normal ground atmospheric pressure
(d)	Attitude	Any

4.5 First article tests. The first article sample shall be subjected to the following first article tests (4.5.2 through 4.5.2.6). Each Acoustic Locating Device shall be subjected to and satisfactorily pass the acceptance tests and examinations (4.7) both prior to and after completion of the first article tests. When specified in the contract (see 6.2), the contractor shall furnish test reports to the procuring activity. Test reports shall include the following information:

(a) Date of tests and names of person conducting tests

(b) Run or test plan used

(c) Serial number of the Acoustic Locating Device

(d) Test results of first article tests

4.5.1 Test equipment. The following items of test equipment, or equivalent, will be required to perform the tests specified herein:

(a) Accelerometer - Endevco type 2213

(b) Hewlett-Packark universal counter Model 523D

(c) Tektronix oscilloscope Model 535A

(d) Hewlett-Packard pwoer supply Model 721A

(e) Hewlett-Packard audio oscillator Model 200 CD

(f) Stop watch having an accuracy of one-half percent in one minute

(g) Test chamber capable of being pressurized to 1000 psig using salt water (common salt, sodium chloride in tap water, specific gravity 1.025 ±0.015), and must be able to retain the pressure of 1000 psig for a period of five days.

(h) A shock test fixture capable of subjecting the Acoustic Locating Device to shock as specified in 4.5.2.2.

(i) A vibration test fixture capable of subjecting the Acoustic Locating Device to vibrations as specified in 4.5.2.1.

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(j) Fresh water pool capable of accommodating the test geometries and requirements of 4.5.2.6.

(k) Voltmeter (peak reading)

(l) Measurement network (100 Hz to 20 kHz ±0.5 dB response)

(m) Hydrophone

(n) Network analyzer or spectrum analyzer with 100 Hz to 20 kHz response

(o) Exercise Head Mk 85 Mod 1 plus all sealing closures

4.5.2 Environmental tests. The Acoustic Locating Device shall be subjected to the following environmental tests. Unless otherwise specified, inability of of Acoustic Locating Device to pass the acceptance tests and examinations subsequent to the environmental tests consitutes a failure.

4.5.2.1 Vibration. Mount the Acoustic Locating Device (FIGURE 1) as shown in FIGURE 3 and secure it to vibration test fixture. The acoustic Locating Device shall be conditioned at room temperature for a minimum period of 1 hour prior to conducting the resonance surveys. Vibrate the unit for a total of 9 hours (3.4.1 and TABLE I) at room temperature, in accordance with the requirements of Procedure XII of MIL-E-5272.

4.5.2.2 Shock. Mount the Acoustic Locating Device as shown in FIGURE 3 and secure it to the shock test fixture table so that a minimum of 90 percent of the table input will be applied to the Acoustic Locating Device under the test. Subject the Acoustic Locating Device to the shocks listed in TABLE III at room temperature using the procedure specified in Shock Test, Procedure IV, of MIL-E-5272. Shock magnitude and duration shall be measured by an accelerometer mounted on the input table of the fixture to indicate applied shock. Test equipment is not restricted to that specified in MIL-E-5272.

TABLE III. Shock schedule.

Peak magnitude (g's)	Minimum duration (ms)	Impact direction	Total No. of impacts
300	0.5	Tranverse plane	3
85	0.5	Verticle axis	6
45	20.0	Transverse plane	3
60	8.0	Any	9

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4.5.2.3 Acceleration. The Acoustic Locating Device shall be mounted as shown in FIGURE 3 and subjected to the acceleration magnitudes and directions required by 3.4.3 for six minutes at room temperature.

4.5.2.4 Temperature, nonoperating. Place the Acoustic Locating Device, preserved and packaged in accordance with 5.1.1, in a temperature controlled chamber and subject it to the following temperature cycle:

(a) Cool to -45, +0 -5 degrees F within 30 minutes.

(b) Maintain at -45, +0 -5 degrees F for 4 hours

(c) Heat to 140, +5 -0 degrees F within 30 minutes.

(d) Maintain at 140, +5 -0 degrees F for 4 hours.

(e) Return to room temperature within 15 minutes. Repeat this schedule for a minimum of three cycles.

4.5.2.5 Operational service life. The operational service life test shall determine that the Acoustic Locating Device shall operate submerged in salt water for five days (see 3.5.1).

(a) Install the Acoustic Locating Device on a suitable fixture.

(b) Attach the accelerometer (4.5.1(a)) to the 1-14NS stud.

(c) Activate the Acoustic Locating Device by immersing in the unpressurized test chamber (4.5.1(g)). Do not allow the accelerometer or the associated circuitry to contact the salt water.

(d) Perform the operational performance test of 4.7.1.2(b) through (f). The peak output voltage amplutide measured shall be equal to the amplitude measured previously in the tests of 4.7.1.2(f).

(e) Remove the accelerometer, and with the Acoustic Locating Device placed in the test chamber (4.5.1(g)) pressurize the chamber to 1000 ± 100 psig.

(f) Maintain the chamber pressure for five days.

(g) Depressurize the chamber and repeat the measurement of 4.5.2.5(b) through (d). The peak output voltage amplitude shall not have decreased by more than 6 dB from the initial measurement taken before pressurization.

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4.5.2.6 Output signal and output level test. The Acoustic Locating Device shall be mounted in the Mk 85 Mod 1 Exercise Head shell (2539065) in accordance with OP 3389. The shell shall be sealed using end bells and necessary closures. The output level and output signal shall be measured with the exercise head shell centerline oriented vertically. Measurements shall be taken in the transverse plane as specified in 3.3.3. The sealed exercise head shall be lowered into a fresh water pool (4.5.1(j)) and oriented as shown in FIGURE 4. Free measurement time on the pulse shall be 10 msec minimum.

a. The Acoustic Locating Device shall be activated by shorting pins D and E of the external electrical connector before installing in the exercise head. The shorting bar shall present a maximum resistance of one ohm between pins D and E.

b. Attach end bells in accordance with OP 3389 paragraph 8-264 (steps 11 and 12) and paragraph 8-265 (steps 5 and 6).

C. As defined in FIGURE 4 at $\theta=0°$, using the test setup of FIGURE 5, measure the pulse length, pulse repetition rate and frequency. The pulse length shall be 30 ±5 msec measured between the 90 percent rise and 10 percent fall points of the pulse. The pulse repetition rate over a period of 60 ±0.5 seconds shall be 60 ±6 per minute. The frequency of the highest amplitude spectral component of the pulse signal shall be 9.000 ±0.270 kHz.

d. Using the test setup of FIGURE 6, measure the sound pressure level through 360 degrees of the transverse plane. This corresponds with $\theta=0°$ to 360° in FIGURE 6. The sound pressure level for all θ, when θE (-20°, +20°) shall be 149 dB minimum referred to one micropascal at one meter.

4.6 Periodic production tests. The periodic production sample, selected in accordance with 4.3.2, shall be subjected to the following tests (4.6.2). When specified in the contract (see 6.2), the contractor shall furnish test reports to the procuring activity. Test reports shall include the following information:

(a) Items 4.5(A) through 4.5(c)

(b) Test results of periodic production tests (4.6.2(a), (b), and (c)

4.6.1 Periodic production test equipment. The following items of test equipment are required to perform the periodic production tests specified herein:

(a) Shock test fixture (see 4.5.1(h))

(b) Vibration test fixture (see 4.5.1(i))

(c) Acceptance test equipment (see 4.7)

(d) Output signal and output level test equipment (see 4.5.1(j) through (m)).

4.6.2 Periodic production test schedule. Subject the Acoustic Locating Device to the following tests in the order indicated:

(a) Acceptance tests and examinations (4.7.1)

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(b) Vibration (4.5.2.1)

(c) Shock (4.5.2.2)

(d) Output signal and output level test (4.5.2.6)

(e) Acceptance tests and examination (4.7.1)

4.7 Acceptance test equpment. The following items of test equipment are required to perform the acceptance tests specified herein:

(a) Counter (see 4.5.1(b))

(b) Oscilloscope (see 4.5.1(c))

(c) Accelerometer (see 4.5.1(a))

(d) Audio Oscillator (see 4.5.1(e))

(e) Stop watch (see 4.5.1(f))

4.7.1 Acceptance tests and examinations. The following tests and examinations shall be conducted on each Acoustic Locating Device to be delivered. Failure to pass any test or examination specified herein shall be considered cause for rejection of the unit.

(a) Operational performance test (4.7.1.2)

(b) Acceptance examination (4.7.1.3)

4.7.1.1 Operational test setup (see FIGURE 2, Test Connections).

(a) Attach the accelerometer (4.5.1(a)) to the 1-14 NS stud.

(b) Connect the output of the accelerometer (4.5.1(a)) to the Oscilloscope (4.5.1(c)).

(c) Connect the audio oscillator (4.5.1(e)) to the external sweep input terminals of the oscilloscope (4.5.1(c)).

(d) Set the horizontal sweep selector control to EXTERNAL.

(e) Connect the counter (4.5.1(b)) to the output of the audio oscillator (4.5.1(e)).

(f) Adjust the counter (4.5.1(b)) for a frequency measurement.

4.7.1.2 Operational performance test. With the Acoustic Locating Device connected to test equipment as in 4.7.1.1, conduct the following test and record results:

(a) Connect a jumper wire from pin D to pin E of the electrical connection.

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(b) Using the stop watch (4.5.1(f)), count pulses for 60 seconds. The number of pulses counted shall be between 54 and 66 to verify the repetition rate of 1.0 ±0.1 per second (3.3.2).

(c) Adjust the audio oscillator frequency (4.5.1(e)) until a clearly defined line, ellipse, or circle appears on the oscilloscope (4.5.1(c)).

(d) Record oscillator frequency. Frequency shall be between 9.270 and 8.730 kcps to verify frequency of 9.000 ± 0.270 kcps (3.3.2).

(e) Set scope horizontal sweep to INTERNAL 5 ms/cm and record pulse length. Pulse length shall be 30 ± 5 ms.

(f) Measure the peak output voltage of the accelerometer and record the result.

4.7.1.3 Examination of product. The Acoustic Locating Device shall be examined for conformance to the requirements specified herein and on Drawing 2158843 or 3277485. The presence of any one of the following defects shall constitute a failure.

(a) Visible holes, cracks, or perforation in housing

(b) Component parts not new; evidence of reclaimed, used or salvaged components

(c) Unit configuration extends beyond drawing limits; dimensions do not comply with drawing requirements

(d) Evidence of unauthorized material and repairs

(e) Identification or polarity of battery terminals incorrect

(f) Threads not in accordance with drawing requirements and FED-STD-H28

(g) Unit weight exceed 3.5 lbs.

(h) Identification and markings not located as specified; not legible or data incomplete

(i) Evidence of damage to housing; scratches or small dents

(j) Preservation, packing, or packaging of the Acoustic Locating Device is not as specified

5. PACKAGING

5.1 Preservation and packaging. Preservation and packing shall be level A or C as specified (see 6.4).

5.1.1 Level A. Cleaning, drying, preservation, and packaging methods for level A shipment shall be in accordance with MIL-P-116, Process C-1 and Method 1A-8.

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5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, preservation, and packing shall be in accordance with the supplier's commercial practice, when such practices have been proven by testing to conform to all applicable requirements.

5.2 Packing.

5.2.1 Level A. The Acoustic Locating Device, preserved and packaged in accordance with 5.1.1, shall be packed in a container in accordance with MIL-P-116, Method IA-2.

5.2.2 Level B. Not applicable

5.2.3 Level C. The Acoustic Locating Device, packaged as specified in 5.1.3, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Normal marking. In addition the markings required by the congract or order, unit packaged and shipping containers shall be marked in accordance with the requirement of MIL-STD-129.

6. NOTES

6.1 Description and intended use.

6.1.1 Description. The Acoustic Locating Device is a self-contained unit which can function independently of any other system. The Acoustic Locating Device is boss mounted and produces a pulsed acoustic signal to the case. The Acoustic Locating Device remains in a static, ready condition, until activated. Activation is effected by connecting two pins of the external connector. The output is pulsed, providing a signal for a minimum period of five days.

6.1.2 Intended use. The Acoustic Locating Device is intended for use in the exercise head of the Torpedo Mk 46 as a signaling device to aid in locating the Torpedo 46 Mk 46 after an exercise or proofing run.

6.2 Ordering data. Procurement documents should specify the following:

(a) Specification control drawing number

(b) Title, number and date of this specification

(c) Serialization requirements

(d) Requirement for a first article (see 3.1, 6.5)

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(e) Assigned activity for first article and periodic tests (see 4.3.1, 4.3.2)

(f) Whether test reports are required (see 4.5 and 4.6)

(g) All additional samples reqired shall be provided and inspected at the expense of the supplier, when such additional samples are required because of failure of any Acoustic Locating Device to pass the prescribed periodic production inspections.

(h) Required level of packing (see 5.2)

(i) Any exceptions to this specification

6.2.2 Data requirements. When this specification is used in an acquisition which incorporate a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirements	Applicable DID no.
4.5., 4.6	Reports, Test	DI-T-2072

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Definitions.

6.3.1 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or to acceleration. The displacement is given in inches peak-to-peak. The acceleration is given in multiples of gravitational acceleration (g's). The frequency is given in cycles per second (cps).

6.3.2 Acoustic Locating Device conditions.

6.3.2.1 Nonoperating. The condition of the uninstalled Acoustic Locating Device when stored, transported, or handled.

6.3.2.2 Ready, nonoperating. The condition of the Acoustic Locating Device when assembled with the end item, transported and handled.

6.3.2.3 Ready. The condition of the Acoustic Locating Device when fully assembled with the end item and installed on the launching mechanism prior to operating.

6.3.2.4 Operating. Operating is the condition of the Acoustic Locating Device when operating.

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6.3.3 Shock. Shock is an abrupt change in the motion of body resulting from the sudden application of a large external force of short duration. Shock is described as a pulse in multiples of gravitational acceleration (g's), duration in units of time (milliseconds (ms)), and a shock pulse approximating a one-half sine wave. The time duration shall be measured betwween two points where two lines, tangent at maximum slope, intersect the zero "g" line.

6.3.4 Acceleration (steady state). Acceleration (steady state) is the time rate of change of velocity of a body resulting from the application of a sustained external force with a uniform build-up to peak in more than 0.010 second. Acceleration is given in multiples of gravitational acceleration (g's).

6.3.5 Temperature. Temperature is the temperature of the environment to which the Acoustic Locating Device is exposed as expressed in degrees Fahrenheit (°F).

6.3.6 Pressure. Unless otherwise specified, pressure is the pressure of the environment to which the Acoustic Locating Device is exposed, expressed in pounds per square inch absolute (psia) or pounds per square inch gage (psig).

6.4 Preparation of delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

6.4.1 Level A. This level shall be for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

6.4.2 Level B. Not applicable.

6.4.3 Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to a domestic installation for immediate use at the first receiving activity.

6.5 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of two units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations and test approval of the documents first article.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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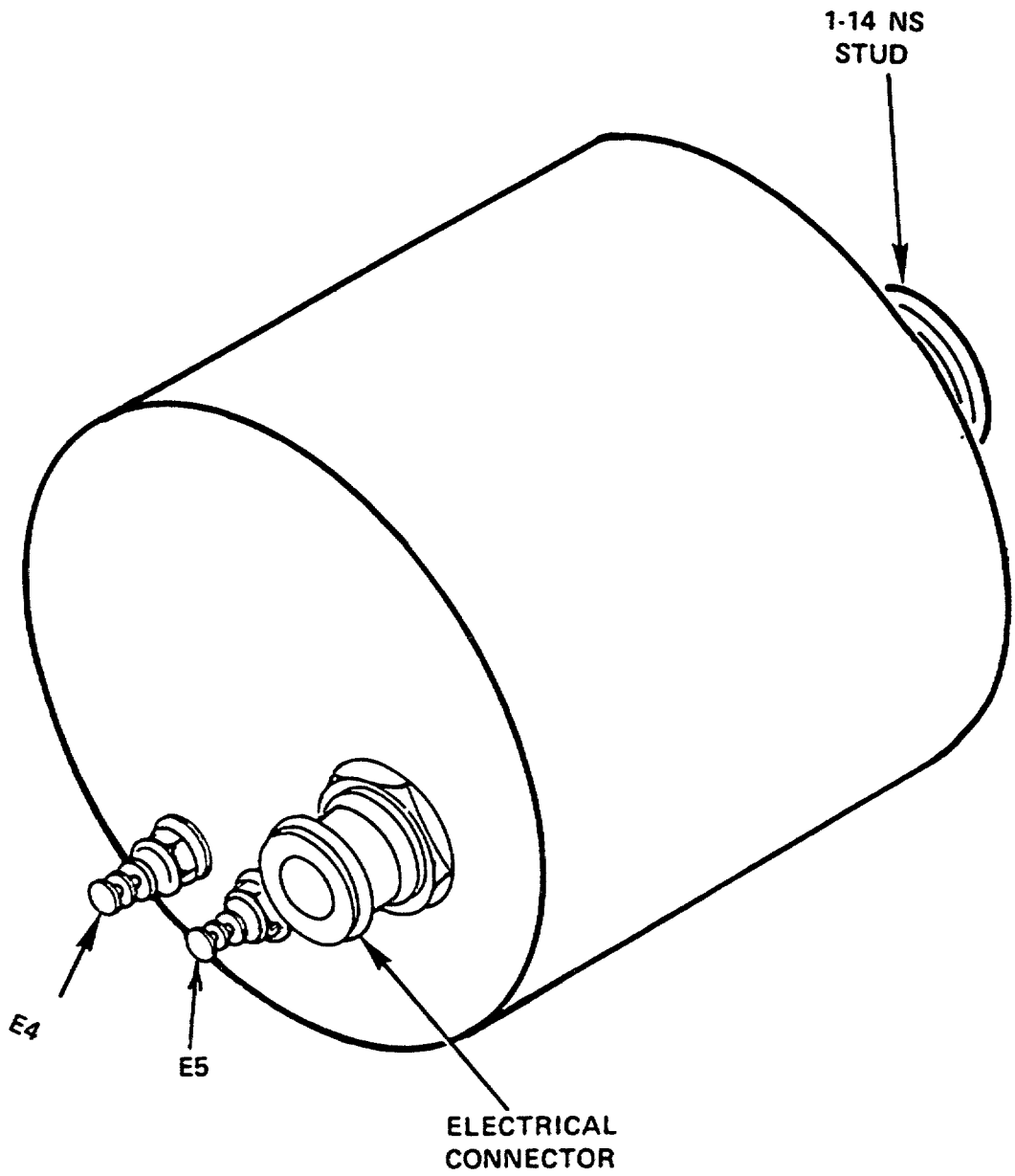


FIGURE 1. Acoustic locating device.

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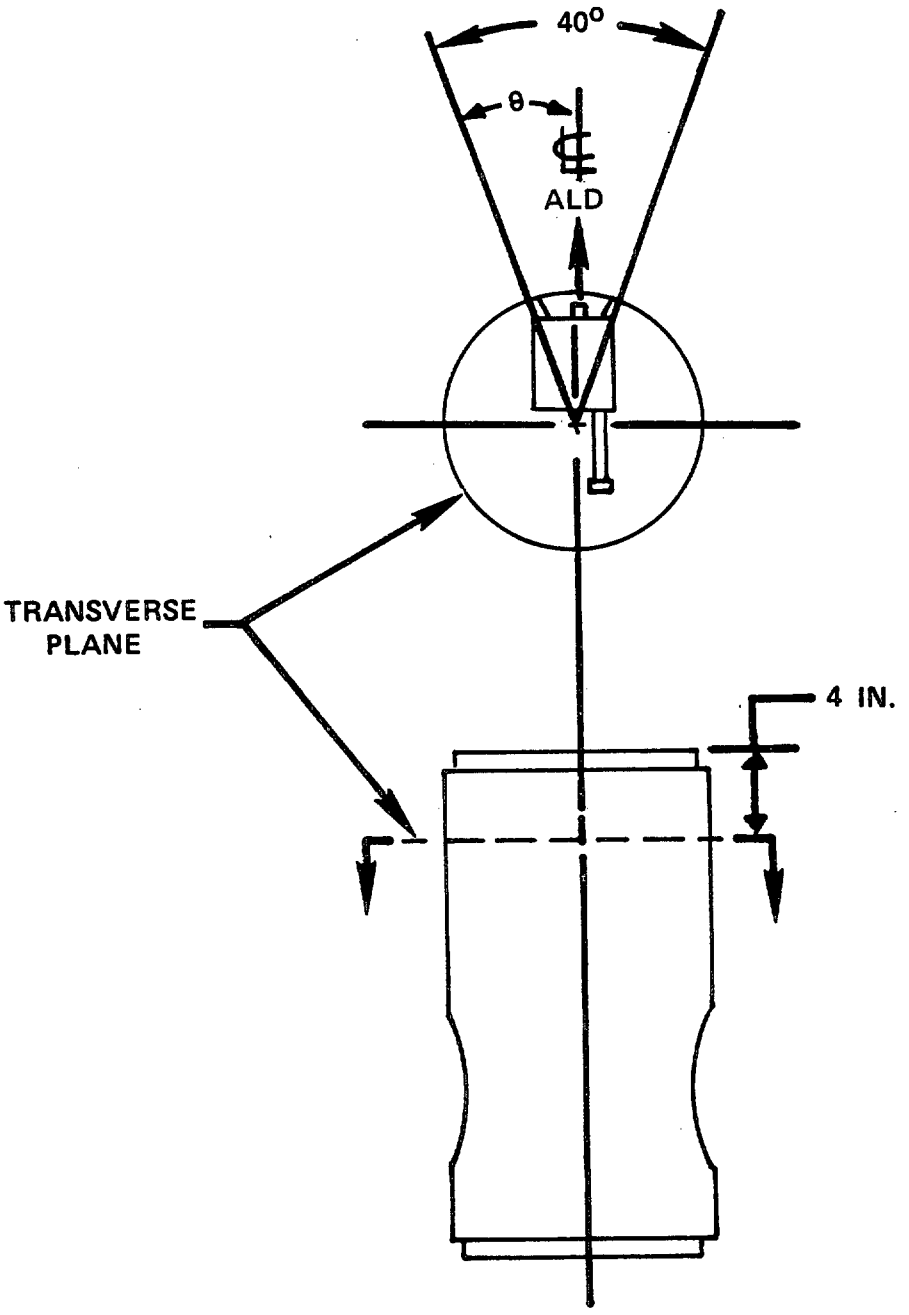


FIGURE 2. Transverse plane location and arc description.

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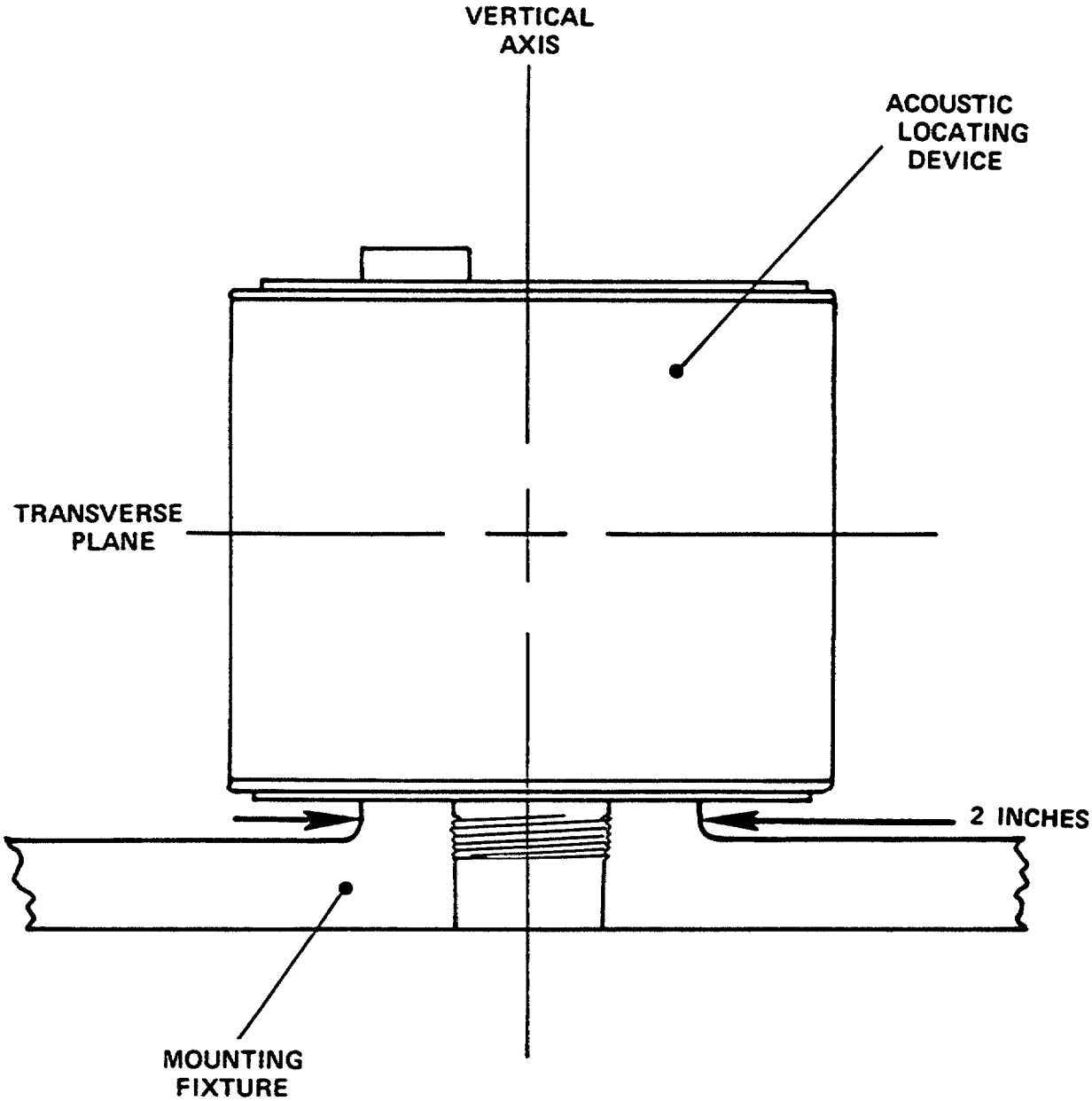


FIGURE 3. Axes of the acoustic locating device.

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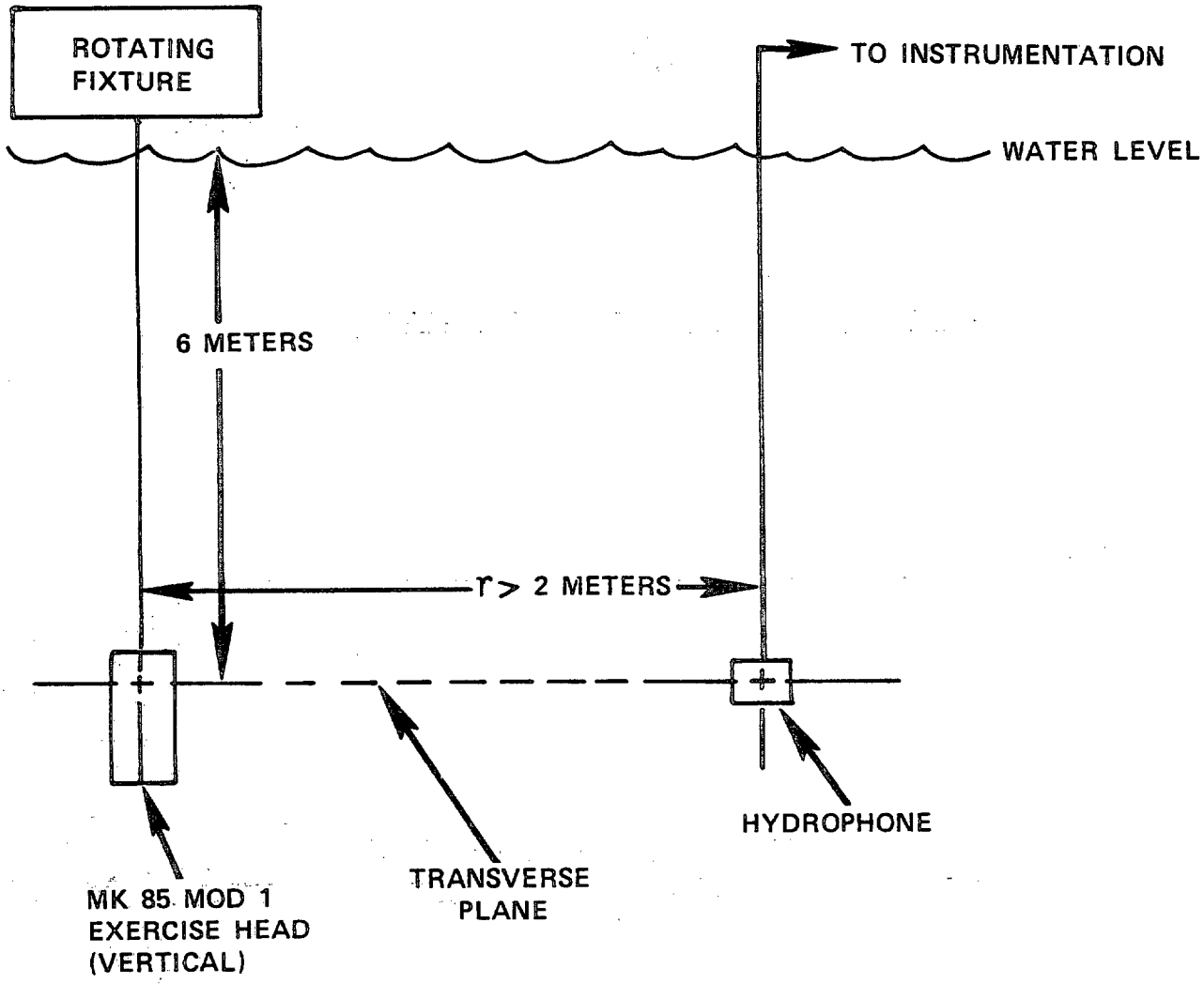


FIGURE 4. Orientation of Mk 85 MOD 1 Exercise Head

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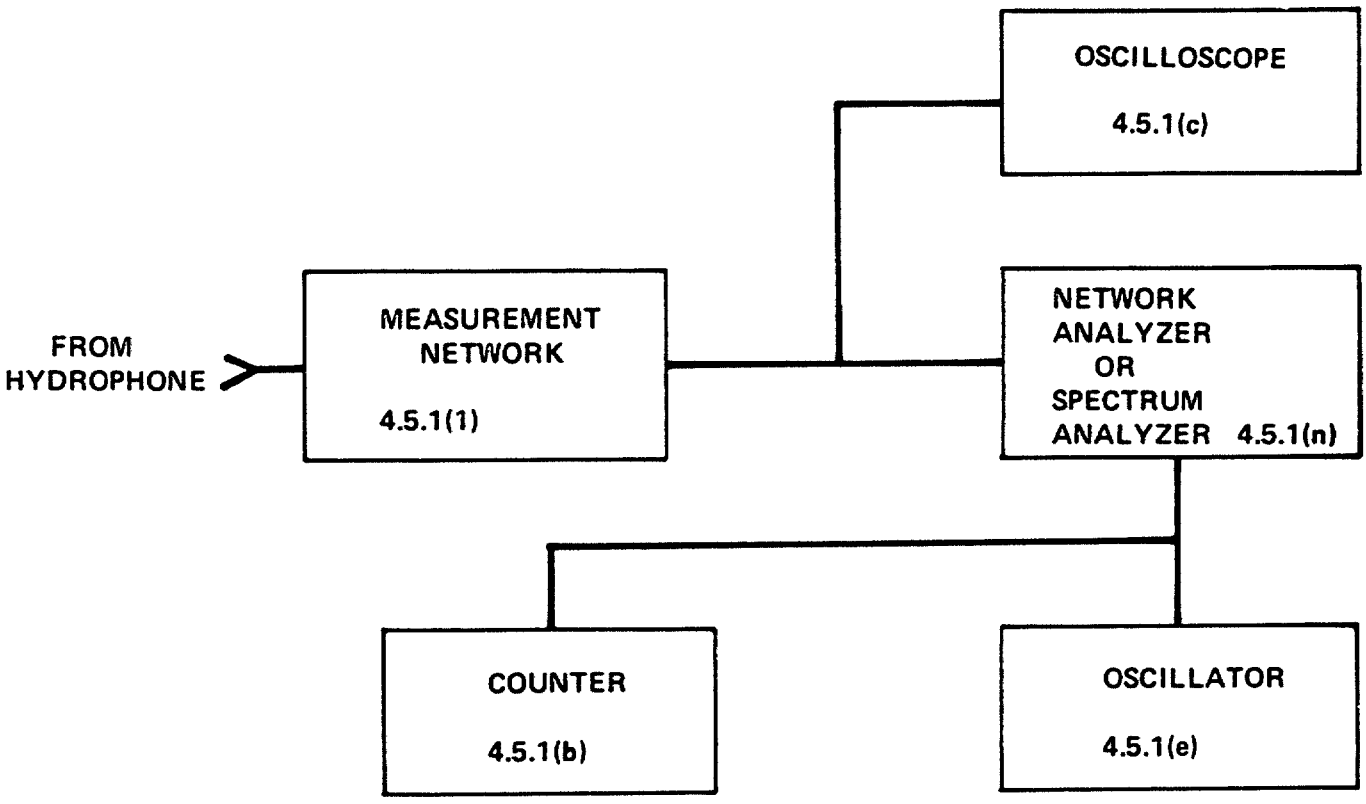
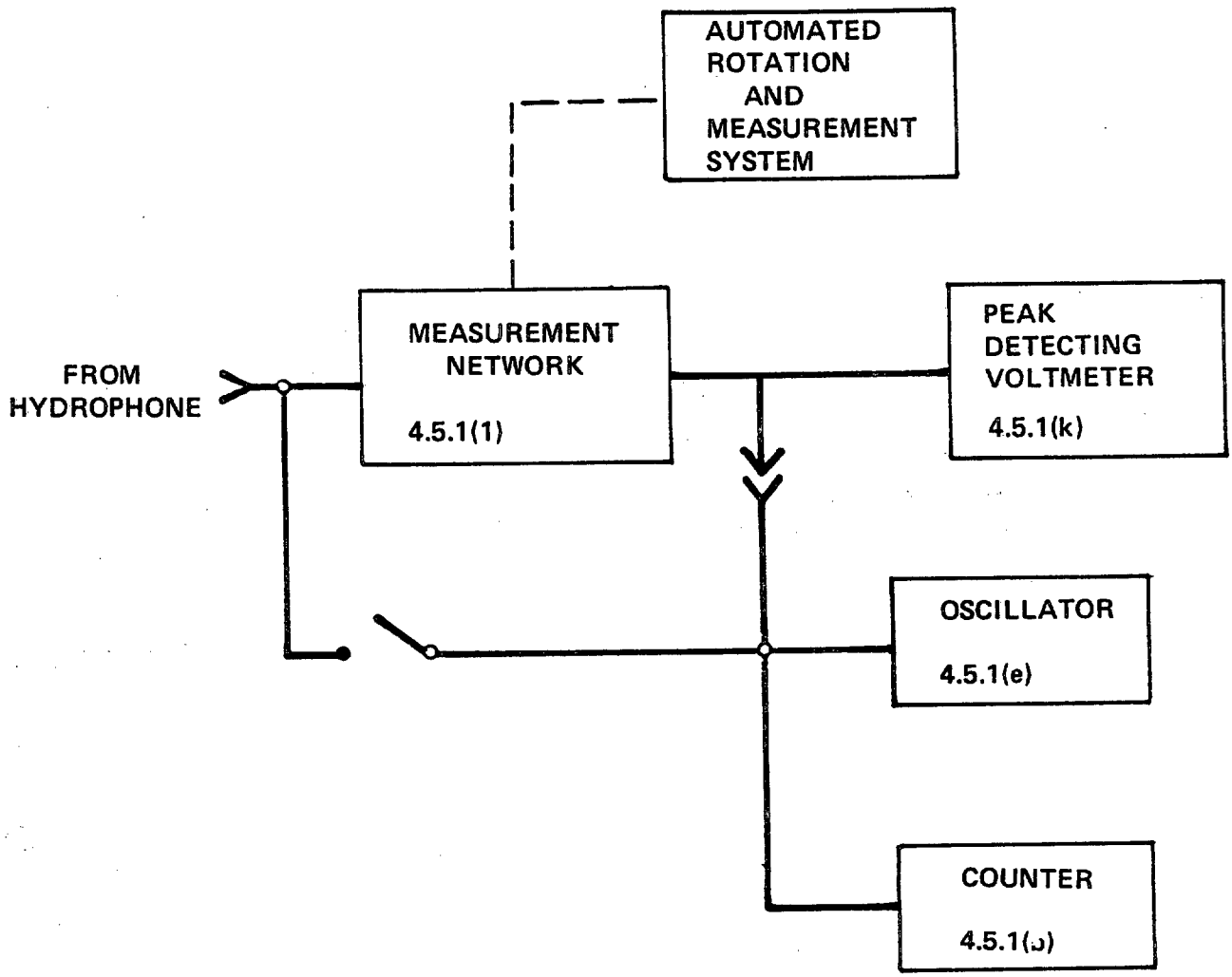


FIGURE 5. Pulse test setup.

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NOTE: (1) Automated rotation and measurement equipment is optional.
(2) Total system measurement accuracy shall be ± 1 dB referred to a micropascal at 1 meter.

FIGURE 6. Sound pressure test setup.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-81327A(OS)	ACOUSTIC LOCATING DEVICE (For Torpedo Mk 46 Mods)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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